FHIR-Based PVC Data Ingestion

I. Clinical context

This script ingests premature ventricular contraction (PVC) burden information into FHIR R4. Although PVCs may be benign, a persistently elevated frequency can indicate increased arrhythmic risk. The metric is encoded with LOINC `8616-5` as an hourly rate (`PVC/h`) in `Observation.valueQuantity`. Literature [2], [1] helped us define clinically relevant thresholds: low burden (1–5 PVC/h) and high (critical) burden (45–50 PVC/h).

II. Code structure and behavior

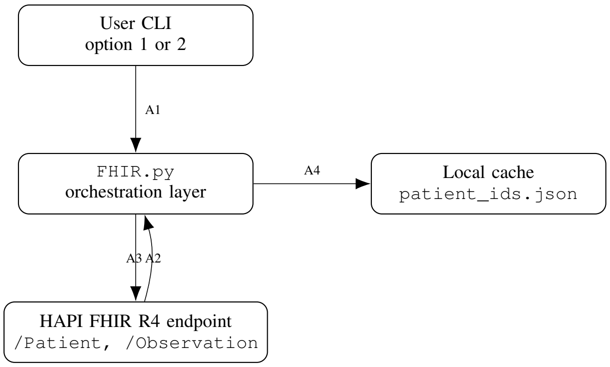
`src/FHIR.py` implements a pipeline architecture that separates persistence, transport, and payload construction. The persistence layer (`load_patient_ids`, `save_patient_ids`) maintains a local map from business identifiers to server logical IDs, enabling resumable runs. The transport layer (`post_resource`, `get_resources`) centralizes HTTP access to the HAPI FHIR server[1] using FHIR media types. Query helpers (`get_patient_by_identifier`, `get_observation_by_identifier`) provide identifier-based retrieval for idempotent interaction.

The resource builders translate internal case dictionaries into standard FHIR resources. Patient demographics are encoded through `Patient.identifier`, `name`, `telecom`, and `address`. PVC measurements are encoded through `Observation` with `status=final`, category `laboratory`, LOINC `8616-5` which corresponds to "premature ventricular contraction" on a rate basis rather than a count (after validation in the official LOINC database we found that there was no specific code for PVC/hour. The other considered code was `76126-2` which is also a code for PVC but expresses a cardinality), patient linkage in `subject.reference`, temporal context in `effectiveDateTime`, and quantified rate in `valueQuantity{value, unit=PVC/h, code=/h}`.

At the workflow level, the CLI exposes `create_patients()` and `send_messages()`. Both first search by identifier and create resources only when absent. Observation uniqueness is enforced by <patientId>-<timestamp>, which reduces duplication on shared servers and improves fault tolerance during interrupted executions.

[1] https://hapi.fhir.org/baseR4

III. Infrastructure and communication



The communication model is intentionally lightweight: one Python client exchanges FHIR JSON with a remote HAPI server and stores only identifier linkage locally. Reliability is achieved through deterministic identifiers, pre-create searches, and cache-backed recovery.

Arrow legend: A1, CLI invocation; A2, FHIR REST calls (GET by identifier, POST create); A3, HTTP response with status and body; A4, Local cache synchronization.

IV. FHIR conformance assessment

The implementation is conformant to HL7 FHIR R4 and satisfies core interoperability requirements: valid resource types (`Patient`, `Observation`), FHIR media types, standard coding systems (`loinc`, `UCUM`, observation-category), and explicit patient–observation references. Identifier search is correctly performed using `[base]/[type]?identifier=system|value`. It does not yet meet full production conformance. Missing elements are the essential (but out of scope) profile-based validation (`meta.profile`/IG constraints), and a clinical-grade security stack with authentication and authorization.

V. Stage scope and conclusion

This stage was not intended to integrate previously developed algorithms in `src/` as a diagnostic engine. The goal was to establish end-to-end HL7 FHIR exchange with a real server and encode clinically meaningful signals for hospital-side interpretation. The prototype highlights core remote-monitoring advantages: interoperable data flow, longitudinal follow-up, earlier detection of adverse patterns, and support for distributed care with fewer unnecessary in-person visits.

References

[1] Hope Cristol and Kim Painter. Premature ventricular contractions (pvc): Symptoms, cause, treatment, 2023. Medically reviewed by Poonam Sachdev, MD. Accessed: 2026-05-14.

[2] Gregory M. Marcus. Evaluation and management of premature ventricular complexes. *Circulation*, 141(17):1404–1418, 2020.